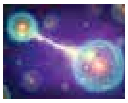
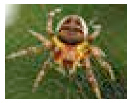
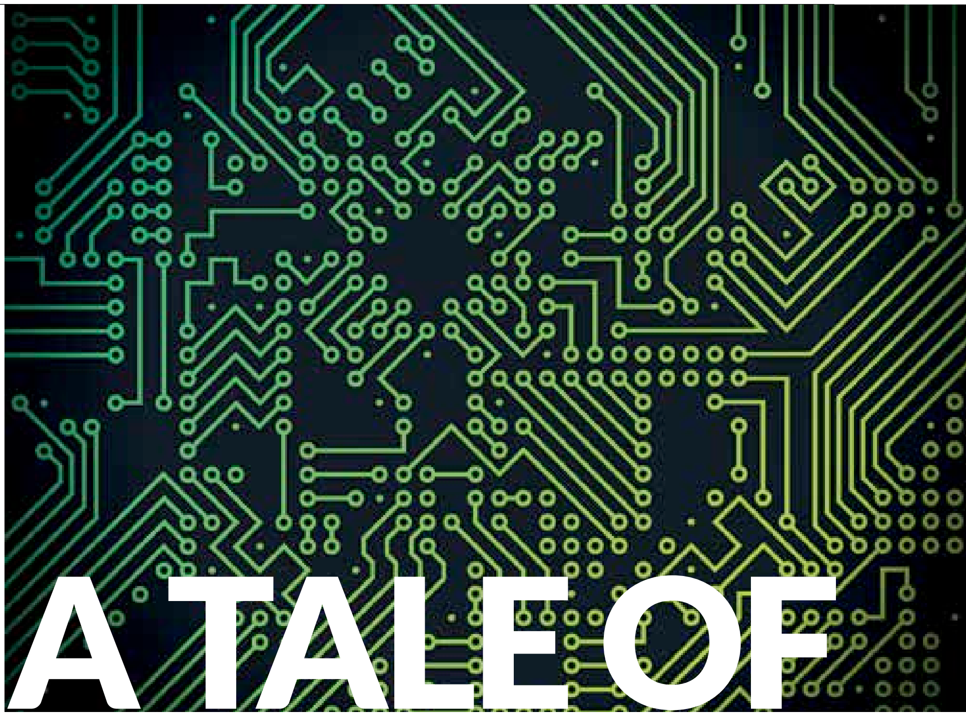


Quantum entanglement record
Chinese physicists have broken a record to establish quantum entanglement across 18 qubits, passing the earlier record of 10 qubits. **https://dgit.in/Aug18-62-1**



Spidey sense is real?
New study suggests that spiders are able to sense the changes in electrical fields in the atmosphere. **https://dgit.in/aug18-62-2**



A TALE OF CHIPS

Could Gallium Nitride be the knight in shining armour in India's semiconductor story?

Arnab Mukherjee | arnab@digit.in

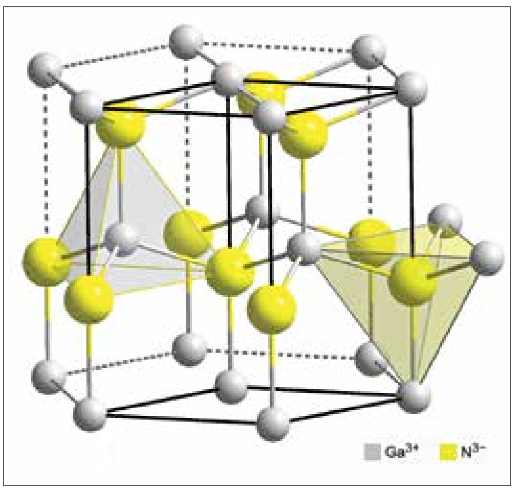
There's no need to establish how crucial semiconductors and their industry are to the world of technology today. Chances are, you are reading this article on a device powered by the same. Even if you're not, if their importance is lost on you, then you're in the wrong place.

Currently, China has nearly 60% hold on the world's semiconductor business with that number poised to grow in the coming years. Even after China, there's a long list of regions like the Americas, Europe, Japan after which there might be a spot for India. On the other hand, we tend to mostly import our electronics or manufacture a large quantity of electronics devices based on imported semiconductors. According to a NOVONOUS report, the consumption of semiconductors in India, mostly import-based, is estimated to rise from $10.02 billion in 2013 to $52.58 billion by 2020 at a dynamic CAGR of 26.72%. Our strength lies in chip design, which our lacking manufacturing capabilities have failed to capitalise on – yet.

A NEW HOPE

Before we begin, we need to segue into our understanding of what semiconductors are made of. Without going into extraneous details, it suffices to say that silicon is an excellent material for building transistors – the things that make your processors run – and is generally what they're made of. Now, silicon transistors are generally good at working under 400 volts. When you switch over the choice of material to gallium nitride, however, the operating threshold moves up from 400 volts to 900 volts. Additionally, gallium nitride has a lot of usefulness when it comes to power electronics.

It is exactly this that Mr. Srinivasan Raghavan had in mind when he moved back to India more than a decade ago. When he arrived here, though, he found a country lacking the expertise to produce and work with gallium nitride. Why is it important for India to have its own gallium nitride fabrication process? Currently, India is too far behind in the silicon based fabrication game to catch up. Whereas, the gallium nitride process, cheaper than its silicon counterpart, is soon to become widespread in electronics manufacturing.



Molecular structure of Gallium Nitride

For India, the power electronics market currently stands at $36.93billion, and is expected to grow to $51.01billion by 2023 – less than five years from now – with increased usage of gallium nitride devices at the heart of this growth. So, to achieve India's goal of zero trade deficit for electronics, which currently stands higher than gold and lower than